

教學實作：擴充 DHT11 溫濕度感測器功能

AMB82-mini ： PIN 8
HUB 8735 Ultra ： PIN 20
ESP32-S3 ： PIN 21

步驟 1：修改硬體定義提示詞 (Devices Definition)

讓 AI 認識 DHT11 感測器及其物理特性。

- DHT11 Temperature & Humidity Sensor
- Pin mapping: depends on development board
 - AMB82-mini: PIN 8
 - HUB 8735 Ultra: PIN 20
 - Measures: temperature (°C) and relative humidity (%)
 - Read mode: single trigger, returns two integer values
 - Temperature range: 0–50 °C
 - Humidity range: 20–90 % RH
 - Physical Rules: Values are integers. Sensor requires ~1 s between reads.

步驟 2：定義工具 JSON 規範 (Tools Definition)

在 toolsDefinition 中加入 /dht11 指令的格式規範。

Reading the DHT11 temperature and humidity sensor

```
{
  "type": "tool_call",
  "method": "/dht11",
  "params": {
    "pin": "<Device pin number. If the user does not specify a pin, ask first.>"
  }
}
```

Success response:

```
{
  "status": "success",
  "method": "/dht11",
  "temperature": <temperature value>,
  "humidity": <humidity value>,
  "workId": "<system-provided>"
}
```

Error response:

```
{
  "status": "error",
  "method": "/dht11",
  "reason": "<error reason>",
  "workId": "<system-provided>"
}
```

步驟 3：底層 C++ 程式碼實作

在專案中新增 `tool_dht11` 驅動函數。

```
#include "DHT.h"
```

```
String tool_dht11(int pin, String workId) {  
    DHT dht(pin, DHT11);  
    dht.begin();  
    float h = dht.readHumidity();  
    float t = dht.readTemperature();  
  
    if (isnan(h) || isnan(t)) {  
        return "{\"status\":\"error\","  
                "\"method\":\"/dht11\","  
                "\"reason\":\"dht11_read_failed\","  
                "\"workId\":\"\" + workId + "\"}";  
    }  
  
    return  
        "{\"status\":\"success\","  
        "\"method\":\"/dht11\","  
        "\"temperature\":\"" + String(t) + \","  
        "\"humidity\":\"" + String(h) + \","  
        "\"workId\":\"\" + workId + "\"}";  
}
```

步驟 4：更新工具分發器邏輯 (executeTool)

將 /dht11 方法註冊進系統的分發流程。

```
void executeTool(String workId, String command, JsonObject params, bool reCheck = true) {

    if (command == "/digitalwrite" || command == "/analogwrite") {
        // ... 原有內容不變 ...

    }

    else if (command == "/dht11") {
        int pin = params["pin"].as<int>();

        String response = tool_dht11(pin, workId);

        if (xSemaphoreTake(stateMutex, MUTEX_TIMEOUT_TICKS) == pdTRUE) {
            historicalMessages += buildGeminiMessage("user", command + timestamps);
            historicalMessages += buildGeminiMessage("model", response + timestamps);
            executeToolHistory += workId + " " + command + " [ " + response + " ]\n";
            xSemaphoreGive(stateMutex);
        }

        evaluateWorkflowContinuation(workId, reCheck);

    }

    else if (command == "/digitalread" || command == "/analogread") {
        // ... 原有內容不變 ...
    }
    // ... 其餘 tools 不變 ...
}
```